Table 15.9 Relative contribution of sources of radiation to the population (UK National Radiation Protection Board, 1989); annual average dose is 2.2 mSv

Natural sources	87
radon	47%
food and drink	12%
γ's from ground and buildings	14%
γ's from cosmic rays	10%
Artificial sources	13%
medical	12%
fall-out	0.4%
work	0.2%
occupational[†]	<0.1%
others	0.4%

[†]Sellafield reprocessing plant, average 5 mSv/y

make the largest contribution to the present radiation background. Typical indoor values are 50—400 Bq/m^3 air; we discuss this further in §15.13.5.

15.11. SOMATIC EFFECTS OF LOW RADIATION DOSES

The effect of low radiation doses such as the natural radiation background or from the nuclear power fuel cycle (and possible accidents) is controversial because the evidence is inconclusive.

15.11.1. Epidemiological results

Of the large number of epidemiological investigations on the effects of low-LET low-level radiations, we will only mention a few representative ones.

Figure 15.9 shows the malignant mortality rate of the US white population accumulated over 18 years (1950—1967) in 46 states as a function of the natural radiation background. Each point represents an average of about 100 000 deaths and is thus significant. The average background value for the whole US population is 1.3mSv/y and the average annual mortality rate in cancer (horizontal line) corresponds to 0.15%. The only significant trend seems to be a decrease in cancer deaths with increasing background up to about 2.5 mSv/y. Although the background radiation in the 7 highest states is 2.1 mSv/y as compared to 1.3 mSv/y for the US average, the frequency of all malignancies is lower (126 per 100 000) in the former than in the latter (150 per 100 000) case. The frequency of leukemia, breast cancer, thyroid cancer, and all child malignancies (age 0—9 y) is slightly lower or the same. Although attempts have been made to correlate the cancer decrease rate with other factors, such as living

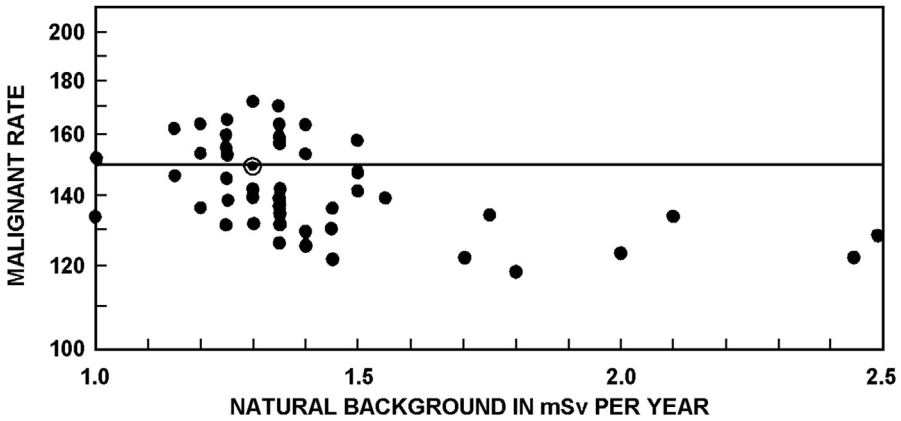


Figure 15.9 Malignant mortality rates per 100 000 by state and natural average radiation background. *(From Frigerio, Eckerman and Stowe, 1973.)*

habits (including sexual frequency and divorce!), no convincing explanation has been given.

This kind of studies are extremely difficult to evaluate since the probability of getting cancer in a country with a high average life span is about 40%. Thus the increased risks due to specific natural radiations will be almost indiscernible.

In the Han province of China, 150 000 peasants with the same genetic background and lifestyle were examined. Half of the group lived in a region where they received almost a threefold higher radiation exposure because of radioactive soil. The investigation of radiation effects such as chromosomal abberations, frequencies of hereditary diseases and deformities, frequency of malignancies, growth and development of children, and status of spontaneous abortions failed to disclose any difference between the two groups.

There are many similar studies from the US, UK, Canada, France, Sweden, Finland, etc, covering millions of people which fail to show a correlation of cancer incidence with small excesses in low-LET radiation. For example, the county in Sweden (Västergötland) with the highest average background radiation has the lowest total cancer frequency.

However, opposite results have also been reported. Kendall et al. (1992; in all 13 authors) have studied the cancer mortality and occupational exposure of 95 000 radiation workers at 24 different sites in England; the mean life-time dose was 34 mSv. They concluded that the frequency of cancers correlated slightly with dose. For multiple myeloma the correlation was "strong" and "for leukemia the trend was significant". 52 leukemia cases were observed while the expected range was 45 ± 13, excluding chronic lymphatic cancer which is assumed not to be related to radiation. Nevertheless, the

observed leukemia frequency was lower than the UK average. The number of all cancers observed was 1363, as compared to an expected 1569 ± 20 in the general population. These "low results" are explained by "the healthy worker syndrome"; i.e. they belonged to a selected group of people with healthy living conditions and good health care and, consequently, the group was assumed not to be representative of the population in general.

A somewhat similar study of Hanford, USA, (see Ch. 19 & 20) workers during 1945–1981 (Gilbert et al. 1989) failed to discover any increase in leukemia, but instead found an increase in multiple myeloma, which also was explained by "the healthy worker syndrome". Multiple myeloma is a skin disease which is common among people who spend much time outdoors and expose themselves excessively to sunshine.

Many studies have tried to relate lung cancer frequency with the Rn-concentrations in ordinary air ($\leq$200 Bq/m^3). Except for the study of uranium miners (§§15.11 and 15.13.5), all these investigations, though carried out with utmost care, have failed to statistically prove such a relation. However, such a relation is still assumed to exist.

15.11.2. Problems in studies of low-level low-LET radiation

In searching for effects of low-level low-LET radiation, the research has focused on the formation of cancer, particulary leukemia, which is a proven effect of large radiation doses, but is a rare form of cancer in the normal population. The normal frequency of leukemia in the US population is 0.012%/year (i.e. 12 cases per 100 000 deaths). The few cases in a large population leads to statistical problems and large uncertainty in the results, as well as difficulties in obtaining a homogenous cohort. Corrections have to be made for sex, aging, changes in the environment (which contain a large number of genotoxic agents), etc. For example, in order to be certain of an increased incidence of breast cancer, which is >10 times more common than leukemia, following an acute exposure of 10 mSv to both breasts at age 35, would require a cohort of 100 million women (Land, Walinder).

Figure 15.10 can be used to illustrate the difficulty with low numbers of incidents. The Figure shows a purely accidental distribution of 165 raindrops on a window. Subdividing the window into 20 squares, A1 to D5 (right figure), one finds a range of 4 (A2) to 13 (D3) drops per square. The difference between the lowest and highest value is over a factor 3! Small excess numbers of cancers, presumably assigned to radiation, carry little weight. Walinder has pointed out that from an epistemological (science theoretical) standpoint, it is impossible to prove that low radiation doses (in the 1 mSv range) are harmful or harmless. As outlined above it is clear that actually trying to prove any effect of ionsing radiation is an epidemiologic challenge. For example in order to ascertain an effect of a dose of 1000 mSv you need a population of 500 persons; 100 mSv need 50000 persons and 10 mSv will require 5 million persons for statistical significance.

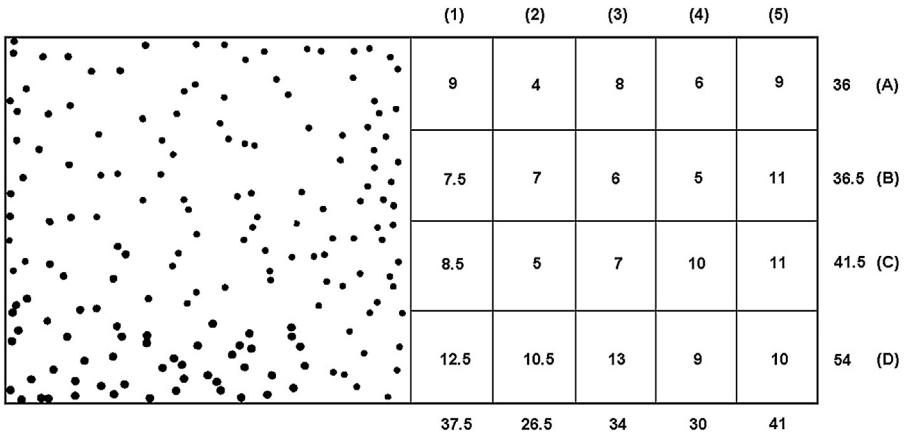


Figure 15.10 Left, distribution of raindrops on a window. Right, matrix, to be put on top of the left Figure. Counting drops in the squares yields the figures noted; error (1σ) is ± 6.

It has been convincingly proven that several damages to the DNA must occur almost simultaneously before a cancerogenic process starts. In §15.2, it was described that low-LET radiation mainly causes single-strand breaks, which are likely to be repaired by the biological system. It was also mentioned that at low dose rates, the radical scavenging system (more generally, the DNA repair system) is more effective than at high dose rates. Further, it was pointed out that at high doses, each cell in a tissue may be hit several times, while at low doses a "hit" cell usually is surrounded by undamaged cells. Low-level low-LET dose effect studies are further complicated by the fact that damage to the *stem cells* in the M phase is the dominating risk. About $0.01-0.1\%$ of all cells are stem cells, which act as breeding sources (especially in the bone marrow) for other cells (which then mature in the thymus or elsewhere for specialized purposes). Thus, research may have to be concentrated on a very small fraction of cells.

Many researchers claim that there are several arguments for the assumption that responses from high-level and high-LET radiation may not be directly extrapolated to low-dose low-LET radiation. Many are of the opinion that the natural radiation background is harmless, and some even claim that it is beneficial (radiation *hormesis*). Thus, looking for effects at low doses can be a vain study. Walinder has pointed out that it is impossible to arrive to a reliable dose-effect relation at doses <50 mSv by epidemiological studies.

A complicating factor in judging the risk of low-LET low-dose radiation is the different sensitivity of the exposed individuals. Though rats etc. can be inbred to produce a single genetic strand, that is not possible with humans. At the beginning of the 20:th century, dozens of young people, mainly female, were employed in the laboratory of Marie Curie making calibrated radium sources, during this work they exposed

themselves to radiation doses of many Sv a year. It is reported – though supressed at that time – that the health of many of these young people rapidly deteriorated so much that they had to leave, some having accured anemia and other radiation related deceases, while others worked for decades without any obvious detrimental health effects. The explanation for this difference is assumed to be due to differences in the individuals immune defence system.

A similar conclusion can be drawn from Figure 15.6.A, where the number of cell abberations – measured a long time after exposure – varies with at least a factor of 5 (i.e. 500% !) for the same dose.

15.12. THE DOSE-EFFECT CURVE

Figure 15.11 shows a number of hypothetical dose-effect relations: The "unmeasurable range" is indicated within the circle. The dashed-dotted line outside the circle indicates the uncertainty in the "measurable range". Line a is based on the by ICRP recommended NTLE (No Threshold Linear Effect) hypothesis and the message is clear: the risk is zero only at zero radiation dose. Curve b indicates a threshold around 50 mSv,

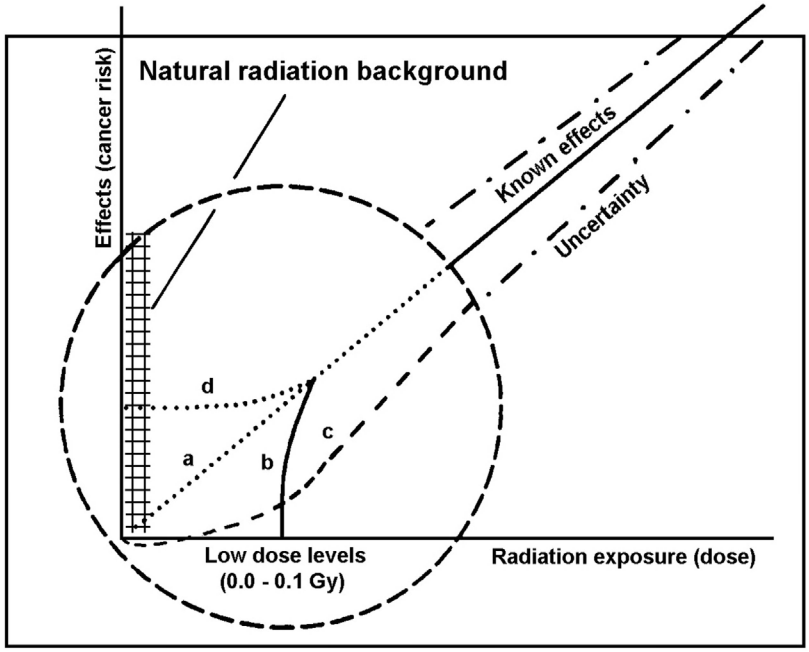


Figure 15.11 Dose effect relations in the known (with spread) and in the unknown (within circle) range, and some hypotheses.